

Gas Field Policy 22: API 618 Comments & Exceptions, 5th Edition (December, 2007)

Siemens Energy builds compressors in basic compliance with this edition of API-618. Our experience has shown that a few of the requirements do not add commensurate life or reliability and therefore the following clarifications are indicated.

SIEMENS ENERGY HOS AND HOSS COMPRESSORS

<u>Paragraph</u>	<u>Comment</u>
6.1.1	The compressors offered are designed and constructed for a minimum service life of 20 years. They are suitable for continuous full load duty for at least 3 years between major overhauls, and should only require replacement of recognized wear components (rings, valves, packing, oil filters, etc.) during this period. Siemens Energy strongly recommends that normal published maintenance schedules be followed.
6.1.3	Unless specifically stated otherwise, capacity may vary by +/- 3%.
6.1.6	If required, Siemens Energy can furnish typical or expected noise level data for this equipment. These noise levels may not comply with the requirements of API 615.
6.1.18	Unless specifically stated otherwise, horsepower may vary by +/- 3%.
6.1.26	Spare/replacement parts...to meet (Par. 9.3.6) except Siemens Energy will not provide specific materials of construction (identified by applicable international standards).
6.2.1	Threads on the HOS/HOSS VVCP piston rod are 2.5"-3 ACME-3G and do not conform to the 8 pitch requirement 1.00" and over diameter.
6.2.3	Internal socket fasteners can be found on the multi-screw tensioners used on piston retainment for the HOS and HOSS compressors. Slotted nut designs have successfully been used for piston retainment on the HOS of decades.
6.2.5	The piston rod for all Siemens Energy HSRC Products is not positively locked to the crosshead assembly. HOS conn rod bolts no longer have a "positive" locking mechanism and now rely only on stretch.
6.3.1	On larger bore cylinders, piston rod vertical run out may exceed the .00015 in./in. of stroke limitation.
6.5.1	The maximum predicted temperature shall not exceed 300°F (adiabatic) and 350°F (actual). In hydrogen rich service, the maximum predicted temperature shall not exceed 275°F (adiabatic) and 300°F (actual).
6.8.1.3	Some of the larger cylinder patterns, when placed side by side, do not allow access to water jacket covers. Some types of unloaders on large cylinders placed side by side are difficult to access.
6.8.1.5	The overbore allowance is 0.062" over the original diameter on all Siemens Energy HSRC Products.
6.8.1.6	Although Siemens Energy's standard for threaded holes in castings is 1-1/2 times the stud diameter, there are a few cases on the HOS and HOSS cylinders which do not meet this criteria.

<u>Paragraph</u>	<u>Comment</u>
6.8.2.1	Some large cylinders require a cylinder support which attaches to the outer head. On packaged units, smaller cylinders are normally supported only from the outer end of the distance piece.
6.8.2.2	All cylinders (except billets) will normally be supplied without liners. Minimum liner thickness is .25" for cylinders up to 9.5" diameter.
6.8.2.3	Finish of typical lubricated compressor cylinder bores is 32RMS.
6.8.3.1	HOS and HOSS cylinders with water jackets are available. All forms of cooling can be accommodated by these cylinder water jackets.
6.8.3.4.1	Refer to 6.8.3.1.
6.8.4.1.10	HOS and HOSS cylinders utilize an extended plug with a hex head to reduce fixed clearances within a given cylinder.
6.8.4.1.12	Siemens Energy only supplies mating flanges and gaskets for the 6000# lens type flanges found on the H.P. billet cylinders.
6.8.4.1.15	Siemens Energy's standard is 90RMS or 100-150 circular finish. All inlet/outlet flanges are ANSI standard with the exception of the HOS/HOSS clamshell cylinders.
6.8.4.2.5	Rectangular flange and bolting dimensions for clamshell cylinders can be found on the cylinder outline drawings provided to the Resellers. Ring type joints will be found on all 4000 psi billet cylinders and lens flanges on the 6000 psi billet cylinders.
6.9.1.3	Siemens Energy valves are not 100% polarized. The valves are designed with a capscrew that protrudes from the valve on the back side of the valve to prevent valves from being installed upside down. In the event a valve was installed upside down, this capscrew would contact the piston during barover inspection and prevent crankshaft rotation alerting operators that the valves are installed improperly. An inlet valve cannot be installed into a discharge valve port because the nose of the valve is too large in diameter. Should a discharge valve inadvertently be installed in a suction valve hole, the valve will not completely seat, so no damage can occur.
6.9.1.5	On a non-conventional cylinder such as the DDV series, the valves are placed, by design, in the cylinder bore.
6.9.1.6	Siemens Energy provides no such valve installation/removal tool.
6.10.2.1	The piston rod is not positively locked to the crosshead on any Siemens Energy HSRC compressor design.
6.10.3.2	HOS and HOSS cylinders are furnished with combination rings, which act as both the wear band and sealing ring. These assemblies are designed for bearing loads under 15 psig. Wear bands and piston rings, designed for the specified bearing load limits, are available as an option.
6.10.4.1	HOS and HOSS cylinders will have the piston rod induction hardened in the area of interest. A tungsten carbide coating (TC3) is available for all Siemens Energy HSRC Products.
6.11.3	The HOS relies on achieving the prescribed bolt stretch to keep the joint tight.

<u>Paragraph</u>	<u>Comment</u>
6.11.4	HOS and HOSS models utilize a box style crosshead of ASTM A536 Grade 65-45-12. This material offers higher strength, durability and better castability than the cast steel. HOS compressors use integral, non-adjustable shoes as standard. Replaceable shim adjustable shoe crossheads are available for HOS models. HOSS models have shim adjustable shoes as standard.
6.12.2.3	The distance pieces are designed for a maximum pressure of 25 psig.
6.12.2.4	On packaged units, a common vent/drain may be utilized at the bottom of the distance piece, if the gas is heavier than air. Vent may be smaller than the 1-1/2" npt specified. Packing gland cooling connections are inlet at the bottom of the distance piece and outlet at the top to avoid formation of air pockets in the packing.
6.12.2.5	External connections may be smaller than the 1" npt specified.
6.12.2.7	In addition to the Austenitic stainless steel tubing and fittings, Siemens Energy may employ a teflon flexible hose with stainless steel braid on pressures below 3000 psi.
6.13.2.3	The HOS/HOSS uses cooled packings for lubricated cylinders with MAWP of 2000 psig or greater as standard. Cooled packings available for all cylinders upon request.
6.14.2.1.1	For pressurized oil systems, temperatures up to 180°F are acceptable provided the proper grade of oil is being used.
6.14.2.1.2	Lubrication system is not built in accordance with API-614.
6.14.2.1.4	j. Standard Siemens Energy relief valves have a cast iron body and hardened steel internals. Any other available combination can be provided at an additional cost. k. HOS and HOSS uses carbon steel upstream of the filter and 300 series stainless tubing with carbon steel fittings downstream.
6.14.2.1.5	Shell and tube coolers utilize cast iron bonnets.
6.14.2.3	b. Siemens Energy standard coolers are not TEMA Class C, but these can be supplied for an additional cost. c. .375 diameter tubes with wall thickness of 22, 24 & 25 BWG and not the 18 BWG as specified.
6.14.2.4	a. All Siemens Energy HSRC Products are 25 micron particle size.
6.14.2.6	Standard Siemens Energy relief valves are located outside the crankcase and have a cast iron body with hardened steel internals. All compressors have their return lines entering the pump suction and do not dump back into the crankcase.
6.14.2.7	Siemens Energy does not supply an oil temperature regulator as standard.
6.14.3.2.2	a. A resettable spring-loaded indicator pin for each outlet is not standard but is available. Instead, Siemens Energy supplies a cycle pin indicator on the distribution block which indicates proper cycling. d. A cycle monitor is not provided.

<u>Paragraph</u>	<u>Comment</u>
6.15.1.11	Siemens Energy does not consider external fasteners or fasteners internal to the distance piece to fall within NACE guidelines. Therefore, Siemens Energy does not require special bolting for these components, however, these items can be supplied if required in the specifications.
6.15.2.1	Siemens Energy reviews cylinder designs with the aid of modern engineering analysis tools to determine cylinder MAWP. Siemens Energy has successfully applied nodular iron as a cylinder material at pressures up to 2750 PSIG (190 bar).
6.15.3.2.1	Siemens Energy nodular iron castings are typically poured to ASTM A536 but can be provided to ASTM A395.
6.15.3.2.2	Charpy V-notch tests are not included as standard.
6.15.3.2.5	Brinell hardness readings are taken at the crankend and headend of the cylinder castings. Readings at other locations are not feasible and are not considered necessary.
6.15.4	Forged steel cylinders are AISI 4132 or AISI 4131 which exceeds the properties of ASTM A668.
6.16.3	Frame nameplates are made from zinc and cylinder nameplates from stainless steel.
6.16.4	Frame nameplates do not include Purchaser's item number or other references. An individual nameplate can be added however, if required.
7.2.1.9	All Siemens Energy HSRC crankshafts are a keyless design.
7.6.6.1	The alarm/shutdown system is the responsibility of the Reseller.
7.7.2.3	The HOS and HOSS uses carbon steel upstream of the filter and 300 series stainless tubing with carbon steel fittings downstream.
7.7.3.1	Internal oil feed lines (when pressures are less than 3000 psi) may be stainless steel braided teflon hose.
7.7.7.2	Vent connections in the packing case and interconnecting tubing will be either a 3/8" stainless steel braided hose, 1/4" - 300 series stainless steel tubing x .065" wall, depending on discharge pressures.
7.11.1	A "tool kit" specifically designed for each of the compressor models is provided in a "ship loose" kit.
7.11.2	b. Siemens Energy does not supply such a valve lifting device. d. Siemens Energy does not supply a crosshead removal and installation tool. e. Siemens Energy HSRC does not have any joints requiring a hydraulic tensioning tool.
7.11.3	Tool boxes are made from heavy gauge Rubbermaid plastic and not steel.
7.11.4	Siemens Energy HSRC do not provide a suitable means of barring compressor over for maintenance unless Siemens Energy is contracted to furnish a flywheel.

<u>Paragraph</u>	<u>Comment</u>
8.3.2.2	a. A helium pressure test is not conducted as standard for cylinders compressing H₂S; only a hydrotest. b. A nitrogen/air pressure test is not conducted as standard for cylinders compressing gases other than those prescribed.
8.4.13	Lifting points only (not center of gravity) are identified on frame outline drawings that are submitted to the Resellers.
8.4.15	Ship loose items such as tools do not have securely affixed, corrosion-resistant metal identification tags indicating items and serial numbers.
9.2.2.1	b. No cross-sectional drawings are provided. c. The only schematic drawings provided are for the cylinder lube oil. d. No sketches are provided, however, the lifting lug locations are noted on the frame and gear outline drawings.
9.2.3.	b. If required, Siemens Energy can furnish typical or expected noise level data for this equipment. These noise levels may not comply with the requirements of API 615.
9.3.1.2	Any submitted drawings will not include date of certification or project identification data.
9.3.1.3	Siemens Energy HSRC does not furnish certified copies of drawings.